PRODUCTION IMPLEMENTATION BLUEPRINT

AI SUPPORT SENTINEL MANUAL

Complete Zero-to-One Deployment, Configuration & Operations Manual for Commercial AI Automation

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Deepak Bagada Systems • Commercial AI Digital Assets

Production Release Standard v1.0.0 • Verified Offline & Zero-Leak Tested

AI Support Triage Sentinel — Implementation Manual

Production Release v1.0.0 | Deepak Bagada Systems

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1. System Overview

The **AI Support Triage Sentinel** is a production AI automation workflow engineered to triage customer support tickets, evaluate churn danger and urgency (1–5 scale), classify categories, draft context-rich replies, and alert your support team via Slack whenever a critical issue or enterprise account needs immediate attention.

Key Performance Metrics

- **Processing Time:** Under 2 seconds per incoming ticket.
- **Fail-Safe Mechanism:** Automatic fallbacks; unhandled errors route to a dedicated admin dead-letter queue.
- **Human-in-the-Loop Safe:** Saves AI responses as *draft notes* so humans retain final control before customer send.

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2. Prerequisites & Credentials Checklist

Ensure you have credentials ready before starting:

Platform	Required Scope	Where to Obtain
n8n or Zapier	Workflow Canvas Access	Self-hosted n8n instance or Zapier account
OpenAI / Claude API	Model execution (`gpt-4o` / `claude-3-5-sonnet`)	platform.openai.com
Helpdesk (Zendesk / Freshdesk)	Webhook Trigger / API Access	Admin $\rightarrow$ Apps & Integrations $\rightarrow$ Webhooks
Slack	Incoming Webhook	api.slack.com/apps

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3. Environment Configuration (`.env`)

Configure the included `.env.example` file:

```
OPENAI_API_KEY=your_openai_api_key_here
SLACK_ESCALATION_WEBHOOK_URL=https://hooks.slack.com/services/YOUR/SLACK/WEBHOOK
PORT=5679
```

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4. Step-by-Step Installation & Deployment

Step 1: Launch the Docker Container Stack

Run this in the product directory to launch the dedicated container stack:

```
docker compose up -d
```

Navigate to `http://localhost:5679` to open your n8n support environment.

Step 2: Import Workflow Blueprint

1. In your n8n interface, click **Workflows $\rightarrow$ Import from File**.

2. Select `workflows/n8n-support-triage.json`.

3. Bind your OpenAI credentials in the **OpenAI GPT-4o Model** node.

4. Bind your Slack Webhook URL in the **Post to #urgent-support Slack** node.

Step 3: Connect Helpdesk Webhook Trigger

1. In Zendesk or Freshdesk, create an automated trigger:

- **Condition:** *Ticket is Created*
- **Action:** *Notify Active Webhook* $\rightarrow$ Target URL: `http://localhost:5679/webhook/support-ticket-inbound`
- **Payload:** Select JSON format matching the schema in Section 5.

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5. Verification & Test Run

Simulate an urgent production ticket by executing this terminal command:

```
curl -X POST "http://localhost:5679/webhook/support-ticket-inbound" \
  -H "Content-Type: application/json" \
  -d '{
    "ticketId": "TCK-99014",
    "customerEmail": "lead-architect@cloudscale.io",
    "planTier": "ENTERPRISE",
    "subject": "CRITICAL: Database connection pool exhausted during peak traffic",
    "description": "Our analytics worker cluster is crashing due to database socket exhausti
  }'
```

Expected Output Response


```
{
  "status": "triaged",
  "ticketId": "TCK-99014",
  "category": "BUG_OUTAGE",
  "urgencyScore": 5,
  "escalated": true,
  "draftedReply": "Hello Cloudscale Team, Thank you for alerting us. Because this involves d
  "timestamp": "2026-10-25T11:35:00.000Z"
}
```


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6. Failure Recovery & Troubleshooting Matrix

Issue	Root Cause	Fix
`400 Bad Request`	Missing required payload keys	Ensure `ticketId`, `subject`, and `description` are included in JSON.
`429 Rate Limit`	High burst of support tickets	n8n will hold and retry using Redis queue backoff.
Slack message fails	Inactive Slack webhook	Re-generate incoming webhook in Slack App Directory.

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